Saddle Nose Deformity

Saddle nose deformity is a characteristic feature of relapsing polychondritis (RP), resulting from inflammation and destruction of the nasal cartilage. This structural collapse can lead to upper airway compromise, obstructive respiratory insufficiency, and predispose to secondary infections. Costochondritis, occurring in approximately 35% of patients, causes parietal chest pain that may further impair respiration due to pain-related splinting.

Musculoskeletal Involvement

Joint pain is a common presenting symptom in RP, affecting about 30% of patients. Involvement of both large and small joints of the peripheral and axial skeleton occurs in over 70% of cases. Arthritis in RP is typically intermittent, migratory, asymmetric, seronegative, and nonerosive.

Ocular Manifestations

Ocular inflammation develops in nearly 60% of patients. The most frequent forms are episcleritis and scleritis, followed by keratoconjunctivitis sicca, iritis, retinopathy, and keratitis. Rare but severe complications—such as corneal perforation, retinal vasculitis, and optic neuritis—can result in blindness.

Audiovestibular Involvement

Conductive hearing loss may arise from stenosis of the external auditory canal, eustachian tube chondritis, or serous otitis media. Sensorineural hearing loss can also occur due to inner ear involvement. Vestibular dysfunction may present acutely with dizziness, ataxia, nausea, and vomiting; symptoms often improve over time.

Cardiovascular Manifestations

Cardiovascular involvement is diverse and potentially life-threatening. It may affect various cardiac structures, including: - Aortic and/or mitral regurgitation - Conduction system abnormalities - Pericarditis

Vascular involvement includes: - Thoracic and abdominal aortitis (may lead to aneurysms) - Takayasu-like aortic arch syndrome - Large and medium vessel vasculitis - Leukocytoclastic vasculitis - Thrombophlebitis

Lesions are primarily inflammatory or thrombotic. Some thrombotic events are associated with anti-phospholipid syndrome, though not all. Large vessel vasculitis often develops after years of subclinical disease activity, even during immunosuppressive therapy.

Dermatologic Manifestations

Skin findings are the initial presentation in about 12% of RP cases and eventually appear in over one-third of patients. These are nonspecific and may include: - Cutaneous nodules on the limbs - Purpura - Oral or complex aphthosis - Papules and sterile pustules - Superficial phlebitis - Livedo reticularis - Limb ulcerations - Distal necrosis

These lesions may occur concurrently with or independently of chondritis episodes. Histopathology reveals inflammatory or thrombotic vascular changes, neutrophilic infiltrates resembling neutrophilic dermatoses, and dermal or subcutaneous inflammation.

While most patients with or without skin involvement have similar systemic manifestations, those with RP associated with myelodysplasia show a higher frequency of dermatologic features (>90%), earlier onset of chondritis, and a skewed male-to-female ratio.

Laboratory Findings

Histopathology

Histopathologic examination of affected cartilage demonstrates perichondrial inflammation and loss of normal basophilic staining of cartilage matrix, reflecting proteoglycan depletion.

Other Laboratory Tests

Laboratory abnormalities in RP are nonspecific and reflect ongoing inflammation—either acute or chronic. Urinalysis may reveal abnormalities if renal involvement is present, such as: - Mesangial expansion - IgA nephropathy - Tubulointerstitial nephritis - Necrotizing glomerulonephritis

Pulmonary function tests, particularly inspiratory and expiratory flow-volume curves, should be performed routinely to detect subclinical respiratory tract involvement. Findings can be confirmed with computed tomography of the airways.

Box 160-1: Empirical Diagnostic Criteria for Relapsing Polychondritis

- Bilateral auricular chondritis
- Non-erosive seronegative inflammatory polyarthritis
- Nasal chondritis
- Ocular inflammation
- Respiratory chondritis

- Audiovestibular damage

Diagnostic Criteria:

A diagnosis of relapsing polychondritis requires one of the following: - At least three of the above clinical criteria

- One or more clinical criteria plus biopsy confirmation of cartilage inflammation

- Chondritis at two or more separate anatomic sites with response to corticosteroids and/or dapsone

Data from Michet CJ et al: Relapsing polychondritis—Survival and predictive role of early disease manifestations. Ann Intern Med 104:74, 1986.